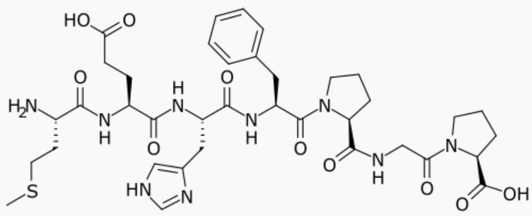
Semax

Semax is a drug which is used mostly in Russia for a broad range of conditions but predominantly for its purported nootropic, neuroprotective, and neurorestorative properties. Semax has not been evaluated, approved for use, or marketed in most other countries. It is prescribed in eastern Europe and Russia for brain trauma.

Semax



Clinical data

Trade names	Semax
Other	L-Methionyl-L-α-

names	glutamylhistidyl-L-phenylalanyl-L-prolylglycyl-L-proline, (Pro8,Gly9,Pro10)ACTH-(4-10)
ATC code	N06BX

Legal status	
Legal status	US: Not FDA approved; unscheduled

Identifiers

IUPAC name

(2*S*)-1-[2-{[(2*S*)-1-[(2*S*)-2-{[2-{[(2*S*)-2-{[(2*S*)-2-amino-4-methylsulfanylbutanoyl]amino}-4-carbo

	xybutanoyl]amino}-3-(1*H*-imidazol-5-yl)propanoyl]amino}-3-phenylpropanoyl]pyrrolidine-2-carbonyl]amino}acetyl]pyrrolidine-2-carboxylic acid
CAS Number	80714-61-0 (https://commonchemistry.cas.org/detail?cas_rn=80714-61-0) ✓ 4037-01-8
PubChem CID	122178 (https://pubchem.ncbi.nlm.nih.gov/compound/122178)
ChemSpider	108969 (https://www

	hemical-Structure.10
UNII	I5FAL2585H (https://precision.fda.gov/uniisearch/srs/unii/I5FAL2585H)
CompTox Dashboard (EPA)	DTXSID601001439 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID601001439)
Chemical and physical data	
Formula	$C_{37}H_{51}N_9O_{10}S$
Molar mass	813.93 $g{\cdot}mol^{-1}$
3D	Interactive image (https://c

model ([JSmol](#))	[du/jmol/jmol.php?model=(BC%40%40H%5D%28C(DO%29C%28N%5BC%4 CC1%3DCNC%3DN1%29 40%40H%5D%28CC2%3 DC2%29C%28N3%5BC% 28CCC3%29C%28NCC% 0%40H%5D%28CCC4% 3DO%29%3DO%29%3D 9%3DO%29%3DO%29% 28CCSC%29N)](#)

SMILES

O=C(N[C@@H](CCC(O)=O)C(
N[C@@H](CC1=CNC=N1)C(
N[C@@H](CC2=CC=CC=C
2)C(N3[C@@H](CCC3)C(NC
C(N4[C@@H](CCC4)C(O)=
O)=O)=O)=O)=O)=O)[C@H](

CCSC)N

InChI

InChI=1S/C37H51N9O10S/c1-57-16-13-24(38)32(50)42-25(11-12-31(48)49)33(51)43-26(18-23-19-39-21-41-23)34(52)44-27(17-22-7-3-2-4-8-22)36(54)46-15-5-9-28(46)35(53)40-20-30(47)45-14-6-10-29(45)37(55)56/h2-4,7-8,19,21,24-29H,5-6,9-18,20,38H2,1H3,(H,39,41)(H,40,53)(H,42,50)(H,43,51)(H,44,52)(H,48,49)(H,55,56)/t24-,25-,26?,27-,28-,29-/m0/s1
Key:AFEHBIGDWIGTEH-CXFOGXNKSA-N

Etymology

Semax is composed of seven amino acid residues: Met-Glu-His-Phe-Pro-Gly-Pro (MEHFPGP), which is reflected in the name - from an abbreviation of "Seven amino acids"- in Russian: СЕМь АминоКиСлот - СЕМАКС.

Marketing

Marketing Development and production: PEPTOGEN (Russian Federation) Joint-Stock Company «Innovative Research-and-Production Center «Peptogen» (JSC «Peptogen»), Russian Federation was founded in 2005 with the participation of the Institute of Molecular Genetics of the Russian Academy of Sciences

Medical uses

Semax 1% from Russia

Semax has undergone extensive study in Russia and is on the Russian *List of Vital & Essential Drugs* approved by the Russian Federation government on December 7, 2011.[1] Medical uses for Semax include treatment of stroke, transient ischemic attack, memory and cognitive disorders, peptic ulcers, optic nerve disease, and to boost the immune system.[2][3][4][5]

Pharmacology

Pharmacodynamics

In animals, Semax rapidly elevates the levels and expression of brain-derived neurotrophic factor (BDNF) and its signaling receptor TrkB in the hippocampus,[6] and rapidly activates serotonergic and dopaminergic brain systems.[7][8] Accordingly, it has been found to produce antidepressant-like and anxiolytic-like effects,[9][10] attenuate the behavioral effects of exposure to chronic stress,[9][10] and potentiate the locomotor activity produced by D-amphetamine.[8][11] As such, it has been suggested that Semax may be effective in the treatment of depression.[12]

Though the exact mechanism of action of Semax is unclear, there is evidence that it may act through melanocortin receptors. Specifically, there is a report of Semax competitively antagonizing the action of the melanocortin receptor full agonist α-melanocyte-stimulating hormone (α-MSH) at the MC_4 and MC_5 receptors in both *in vitro* and *in vivo* experimental conditions, indicating that it may act as an antagonist or partial agonist of these receptors.[13] Semax did not antagonize α-MSH at the MC_3 receptor, though this receptor could still be a target of the drug.[13] As for the MC_1 and MC_2 receptors, they were not assayed.[13] In addition to actions at receptors, Semax, as well as a related peptide drug, Selank, have been found to inhibit enzymes involved in the degradation of enkephalins and other endogenous regulatory peptides (IC_{50} = 10 μM), though the clinical significance of this property is uncertain.[14]

Pharmacokinetics

As a peptide, Semax has poor oral bioavailability and hence is administered parenterally as a nasal spray or subcutaneous injection.

Human trials

In a 2018 study involving 24 healthy participants, Semax was shown to increase fMRI default mode network activity relative to placebo.[15]

In an earlier 1996 study, 250-1000 ug/kg of Semax improved attention and short term memory in 11 healthy subjects performing 8 hour work shifts, though the effects were most pronounced when subjects were fatigued (after the shift was over) and the effects lasted going into the next day.[16] In a follow-up memory test administered the morning after Semax administration, the treatment group made more correct responses (71%) than the control group (41%).[16]

A study involving 110 patients recovering from ischemic stroke reported increases in BDNF (correlated with early rehabilitation) in patients administered Semax.[17]

As of November 2023, there are no published human trials involving Semax outside of Russia and Post-Soviet states.[18]

Chemistry

Semax is a heptapeptide and synthetic analogue of a fragment of adrenocorticotropic hormone (ACTH), ACTH (4-10), of the following amino acid sequence: Met-Glu-His-Phe-Pro-Gly-Pro (`MEHFPGP` in single-letter form).

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